Properties of a Linear Unbiased Minimum Variance Estimator

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Overview

Here is the simplest instance of the problem of this paper. You have n measurements $b_1, \ldots, b_n$ of a single unknown quantity x, with independent errors of differing precisions $\sigma_1^2, \ldots, \sigma_n^2$. Every laboratory scientist knows what to do: combine the measurements with weights proportional to $1/\sigma_i^2$. But why that weight – and in what sense is the result "best"? The answer this paper gives: in *every* reasonable sense at once (see the Special Cases section for the return to this instance). The general setting is as follows.

A standard problem is to find the linear unbiased "minimum" variance estimator for the overdetermined problem[1] : $A\mathbf{x} = \mathbf{b}$, where $A \in \mathbf{R}^{n\times m}$, $\mathbf{b} \in \mathbf{R}^n$ is a random observation with $E[\mathbf{b}] = A\mathbf{x}$ and covariance V (assumed invertible), and $\mathbf{x} \in \mathbf{R}^m$ is the deterministic parameter we wish to estimate. We look for a linear unbiased estimator for $\mathbf{x}$ of the form $\mathbf{x}^* = L\mathbf{b}$, where $L \in \mathbf{R}^{m\times n}$. Since $\mathbf{x}^*$ is unbiased we must have $E[\mathbf{x} - \mathbf{x}^*] = \mathbf{x} - L\,A\mathbf{x} = (I - L\,A)\mathbf{x}$. For this to vanish for every $\mathbf{x}$, we require $L\,A = I$. This implies that A and L have rank m. The covariance of $L\mathbf{b}$ is LVL^T. The problem now is to find the best L such that $L\,A = I$ and the covariance matrix, LVL^T, is minimized in some sense. This is the classical *best linear unbiased estimator* (BLUE) problem; see, e.g., Rao [2].

The problem may be restated as choosing L to "minimize" the norm of LVL^T subject to the constraint $L\,A = I$. The question is what norm does one use. The "natural" norm of a matrix is $\sqrt{\operatorname{tr}(M^TM)}$. This follows from considering the standard basis in $\mathbf{R}^{n\times m}$ and using the usual norm of the vector space $\mathbf{R}^{n\times m}$. The matrix of interest, LVL^T, is positive definite symmetric; in this case, each of the "trace" expressions, $\sqrt[k]{\operatorname{tr}(M^k)}$ $(k \in [1, \infty))$, is a norm for a positive definite matrix $M_{m\times m}$ (these are the *Schatten* k-norms.)

Do the Norms Matter?

The question is, do different choices of norms lead to a different solution, L? To answer this, we solve the constrained minimization problem that results from each of these trace norms[2] :

$$F(L; k) = \operatorname{tr}\left((LVL^T)^k - (L\,A - I)\Lambda_k^T\right) \quad k \in [1, \infty)$$

where $\Lambda_k \in \mathbf{R}^{m\times m}$ is a Lagrange multiplier.[3]

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[1] This means that $n > m$.
[2] Equivalently, we minimize the k^{th} power of the k^{th} trace norm.
[3] For non-integer k, powers of the positive definite matrix LVL^T are defined by the spectral calculus (apply $t \mapsto t^k$ to its eigenvalues); the differentiation rule used below, $d\operatorname{tr}(M^k) = k\operatorname{tr}(M^{k-1}\,dM)$ for symmetric perturbations of a positive definite M, is valid in this setting.

The derivative of F with respect to L is given by:

$$DF(L)(H) = \mathrm{tr}\left[k(LVL^T)^{k-1}(HVL^T + LVH^T) - HA\Lambda_k^T\right]$$

At a minimum value, L^*, the derivative must vanish; therefore,

$$\mathrm{tr}\left[k(L^*VL^{*T})^{k-1}(HVL^{*T} + L^*VH^T) - HA\Lambda_k^T\right] = 0 \quad \forall H \in \mathbf{R}^{m\times n} \tag{1}$$

Before going further we list some facts regarding the trace inner product that will be used in the rest of the paper.

- **Fact 1.** $\forall H \in \mathbf{R}^{n\times n},\ H$ anti-symmetric $\Rightarrow\ \mathrm{tr}(H) = 0$.
- **Fact 2.** Given a matrix $A \in \mathbf{R}^{m\times n}$: $\big(\forall H \in \mathbf{R}^{m\times n},\ \mathrm{tr}(A^TH) = 0\big) \Rightarrow A \equiv 0$.
- **Fact 3.** $\forall A, H \in \mathbf{R}^{n\times n},\ A$ symmetric $\wedge$ H anti-symmetric $\Rightarrow\ \mathrm{tr}(A^TH) = 0$.
- **Fact 4.** Given a matrix $A \in \mathbf{R}^{n\times n}$: $\big(\forall$ anti-symmetric $H \in \mathbf{R}^{n\times n},\ \mathrm{tr}(A^TH) = 0\big) \Rightarrow$ A is symmetric.
- **Fact 5.** Given a *symmetric* matrix $A \in \mathbf{R}^{n\times n}$: $\big(\forall$ symmetric $H \in \mathbf{R}^{n\times n},\ \mathrm{tr}(A^TH) = 0\big) \Rightarrow$ $A \equiv 0$.

The first fact follows from a property of square matrices; namely, $\mathrm{tr}(A) = \mathrm{tr}(A^T)$; and the definition of an anti-symmetric A, $A^T = -A$. The second fact follows since any inner product on a vector space, V, satisfies: $\big(\forall y \in V,\ \langle x, y\rangle = 0\big) \Rightarrow x \equiv 0$. The third fact follows – using Fact 1 (or rather its underlying identity $\mathrm{tr}(M) = \mathrm{tr}(M^T)$) – as follows. If S and W are symmetric and anti-symmetric matrices in $\mathbf{R}^{n\times n}$, then $\mathrm{tr}(S\,W) = \mathrm{tr}((S\,W)^T) = \mathrm{tr}(W^T S^T) = -\mathrm{tr}(W\,S) = -\mathrm{tr}(S\,W)$. This implies that $\mathrm{tr}(S\,W) = 0$. The fourth fact follows from the unique decomposition of any matrix $A \in \mathbf{R}^{n\times n}$ as a sum of a symmetric and an anti-symmetric matrix: $A = A^s + A^w$, so $A^T = A^s - A^w$. For any anti-symmetric H, $\mathrm{tr}(A^TH) = \mathrm{tr}(A^sH) - \mathrm{tr}(A^wH) = -\mathrm{tr}(A^wH)$ by Fact 3. So if $\mathrm{tr}(A^TH) = 0$ for all anti-symmetric H, then $\mathrm{tr}(A^wH) = 0$ for all such H. Setting $H = A^w$ gives $\mathrm{tr}((A^w)^2) = -\mathrm{tr}((A^w)^TA^w) = -\|A^w\|_F^2 = 0$, so $A^w \equiv 0$ and A is symmetric. The fifth fact follows by setting H to A. Again, the fact that the trace inner product induces a norm implies that $A = 0$. (Only Facts 1–3 are used in the derivations below; Facts 4 and 5 are recorded because they round out this family of trace arguments.)

Continuing with the derivation, in the special case $k = 1$, equation (1) implies that $2VL^{*T} = A\Lambda_1^T$; or,

$$2L^{*T} = V^{-1}A\Lambda_1^T \tag{2}$$

Multiplying both sides by A^T and using the fact that $A^TL^{*T} = I$ yields $2I = A^TV^{-1}A\Lambda_1^T$. Since A has rank m, $A^TV^{-1}A$ is invertible, so $\Lambda_1^T = 2(A^TV^{-1}A)^{-1}$ (the right-hand side is symmetric, so $\Lambda_1 = \Lambda_1^T$). Combining this with equation (2) gives

$$L^* = (A^TV^{-1}A)^{-1}A^TV^{-1}$$

We now proceed with the case when $k > 1$. Equation (1) contains the test direction H in two costumes – multiplied against VL^{*T} and against A – so a good choice of H should simplify both occurrences at once. Routing an arbitrary $\tilde{H} \in \mathbf{R}^{m\times m}$ through A^TV^{-1} does exactly that: with $H = \tilde{H}A^TV^{-1}$, the V cancels in the first occurrence (leaving $HVL^{*T} = \tilde{H}A^TL^{*T} = \tilde{H}$) while the second becomes $HA = \tilde{H}\,A^TV^{-1}A$. Then equation (1) becomes:

$$\mathrm{tr}\left[k(L^*VL^{*T})^{k-1}(\tilde{H}A^TL^{*T} + L^*A\tilde{H}^T) - \tilde{H}(A^TV^{-1}A)\Lambda_k^T\right] = 0 \quad \forall \tilde{H} \in \mathbf{R}^{m\times m}$$

Or,

$$\operatorname{tr}\left[k(L^*VL^{*T})^{k-1}(\tilde{H}+\tilde{H}^T)-\tilde{H}(A^TV^{-1}A)\Lambda_k^T\right]=0 \quad \forall\tilde{H}\in\mathbf{R}^{m\times m} \tag{3}$$

since $L^*A=I$ and $A^TL^{*T}=I$.

Equation (3) holds for all $\tilde{H}\in\mathbf{R}^{m\times m}$. Let $M=(L^*VL^{*T})^{k-1}$ and $C=A^TV^{-1}A$. Since M is symmetric, $\operatorname{tr}(M(\tilde{H}+\tilde{H}^T))=2\operatorname{tr}(\tilde{H}M)$ by cyclic invariance, so (3) reads

$$\operatorname{tr}\big[\tilde{H}(2kM-C\Lambda_k^T)\big]=0 \quad \forall\tilde{H}\in\mathbf{R}^{m\times m}.$$

By Fact 2, $2kM-C\Lambda_k^T=0$. Since A has rank m, $(A^TV^{-1}A)^{-1}$ exists, and we can solve for Λ_k^T:

$$\Lambda_k^T=2k\,(A^TV^{-1}A)^{-1}(L^*VL^{*T})^{k-1}.$$

Equation (1) now becomes:

$$\operatorname{tr}\left[k(L^*VL^{*T})^{k-1}(HVL^{*T}+L^*VH^T)-2kHA(A^TV^{-1}A)^{-1}(L^*VL^{*T})^{k-1}\right]=0 \quad \forall H\in\mathbf{R}^{m\times n}$$

Or,

$$\operatorname{tr}\left[\Big((HVL^{*T}+L^*VH^T)/2-HA(A^TV^{-1}A)^{-1}\Big)(L^*VL^{*T})^{k-1}\right]=0 \quad \forall H\in\mathbf{R}^{m\times n}$$

Which may be written

$$\operatorname{tr}\left[\Big(HVL^{*T}+(L^*VH^T-HVL^{*T})/2-HA(A^TV^{-1}A)^{-1}\Big)(L^*VL^{*T})^{k-1}\right]=0 \quad \forall H\in\mathbf{R}^{m\times n}$$

Since $(L^*VH^T-HVL^{*T})/2$ is anti-symmetric and $(L^*VL^{*T})^{k-1}$ is symmetric, the trace of their product is zero. Therefore we may write the last equation as

$$\operatorname{tr}\left[\Big(HVL^{*T}-HA(A^TV^{-1}A)^{-1}\Big)(L^*VL^{*T})^{k-1}\right]=0 \quad \forall H\in\mathbf{R}^{m\times n}$$

Or,

$$\operatorname{tr}\left[H\Big(VL^{*T}-A(A^TV^{-1}A)^{-1}\Big)(L^*VL^{*T})^{k-1}\right]=0 \quad \forall H\in\mathbf{R}^{m\times n}$$

This implies that $\big(VL^{*T}-A(A^TV^{-1}A)^{-1}\big)\,(L^*VL^{*T})^{k-1}=0$. Since L^* has rank m, $(L^*VL^{*T})^{-(k-1)}$ exists, so that $VL^{*T}-A(A^TV^{-1}A)^{-1}=0$. Hence,

$$L^*=(A^TV^{-1}A)^{-1}A^TV^{-1}$$

Two remarks complete the logic, for every $k\in[1,\infty)$. First, a minimizer *exists*: the feasible set $\{L : L\,A=I\}$ is closed and non-empty (it contains L^*), and the objective is coercive on it – since $LVL^T\succeq\lambda_{\min}(V)\,LL^T$ entails, eigenvalue by eigenvalue, $\operatorname{tr}\big((LVL^T)^k\big)\geq\lambda_{\min}(V)^k\operatorname{tr}\big((LL^T)^k\big)$, which grows without bound as $\|L\|\to\infty$. Second, because the constraint is affine, any minimizer must satisfy the stationarity condition

(1). The computations above show that (1) has the *unique* solution L^*; therefore the critical point L^* is in fact the minimizer.

Therefore, it does not matter which of the matrix norms we use: the "best" linear unbiased estimator is the same.

Remark: Connection with the Gauss–Markov Theorem

There is a classical explanation for why the norms cannot matter. The Gauss–Markov theorem, in the generalized form due to Aitken [1], states that L^* minimizes the covariance in the strongest possible sense – in the *partial order* of symmetric matrices: for *every* unbiased L,

$$LVL^T \;\succeq\; L^*VL^{*T},$$

meaning the difference $LVL^T - L^*VL^{*T}$ is positive semi-definite. The proof is short with the machinery at hand. Write $L = L^* + D$; the constraint $L\,A = L^*A = I$ forces $D\,A = 0$. Then the cross term vanishes:

$$L^*VD^T = (A^TV^{-1}A)^{-1}A^TV^{-1}VD^T = (A^TV^{-1}A)^{-1}(D\,A)^T = 0,$$

so that

$$LVL^T = L^*VL^{*T} + DVD^T \;\succeq\; L^*VL^{*T}.$$

The shape of this argument deserves its name: the vanishing cross term says the deviation D is *orthogonal* to the solution in the inner product weighted by V^{-1}, and the conclusion is then Pythagoras. That is, L^* is a *projection* construction – the "best approximation in a subspace" game of least squares. It is the same game played by the pseudo inverse $A^T(AA^T)^{-1}$ of the companion paper on singular transformations of densities – the two pseudo inverses are dual: there, the minimum-norm solution of an *underdetermined* system; here, the least-squares estimate for an *overdetermined* one – and, in function space, by the conditional expectation of probability theory. Every norm considered in this paper – each trace norm, and the maximum eigenvalue used in the next section – is *monotone* in this partial order (by Weyl's eigenvalue monotonicity, $X \succeq Y \succ 0$ implies the eigenvalues of X dominate those of Y one for one); consequently each is minimized at L^*. The variational computations above thus confirm, norm by norm, what the Gauss–Markov inequality delivers all at once; their value is that they *find* L^* rather than merely verify it.

Special Cases: Recovering Known Estimators

Two specializations of L^* recover formulas the reader already owns – which is how one learns to trust a general machine.

Ordinary least squares. If the observation errors are uncorrelated and of equal precision, $V = \sigma^2 I$, then the σ^2's cancel and

$$L^* = (A^TA)^{-1}A^T$$

– the ordinary least squares pseudo inverse. A general V thus enters the estimator exactly as a re-weighting of the data by reliability.

The inverse-variance weighted mean. Return to the instance that opened this paper. Let $m = 1$ with $A = (1, 1, \ldots, 1)^T$ – that is, $\mathbf{b}$ consists of n direct measurements b_i of a single unknown number x – and let

$V = \mathrm{diag}(\sigma_1^2, \ldots, \sigma_n^2)$ (independent errors of differing precisions). Then $A^T V^{-1} A = \sum_i 1/\sigma_i^2$ and

$$x^* = L^* \mathbf{b} = \frac{\sum_{i=1}^n b_i/\sigma_i^2}{\sum_{i=1}^n 1/\sigma_i^2}, \qquad \mathrm{Var}(x^*) = L^* V L^{*T} = \left(\sum_{i=1}^n \frac{1}{\sigma_i^2} \right)^{-1}$$

– the inverse-variance weighted average used to combine measurements of unequal precision. The weight $1/\sigma_i^2$ is not a convention: by the results above it is optimal in every trace norm, in the spectral norm (next section), and – by the Gauss–Markov remark – in the partial order of covariance matrices itself.

The ladder of specializations in one place:

Setting	**Estimator**	**Known as**
$V = \sigma^2 I$	$(A^T A)^{-1} A^T$	ordinary least squares
general V	$(A^T V^{-1} A)^{-1} A^T V^{-1}$	generalized least squares
$m = 1$, $A = (1, \ldots, 1)^T$, V diagonal	weights $1/\sigma_i^2$	inverse-variance mean

Best Estimator under the Spectral Norm

We now use the results of the trace norms above to show that the estimator obtained has the smallest maximum eigenvalue over all linear estimators. The strategy is approximate-then-take-a-limit, played with whole optimization problems as the approximants: the spectral norm is the $k \to \infty$ limit of the Schatten norms, the problem has been solved for *every* finite k, and the lemma below is the license to pass the minimizer through the limit.

Lemma 1. *Let Ω be a closed set in a finite-dimensional space $\mathbf{R}^d$, and suppose the feasible set $\{x \in \Omega : g(x) = 0\}$ is non-empty. If $x^* \in \Omega$ is a common solution to $\min_{\{x \in \Omega : g(x)=0\}} \varphi_k(x)$ for every $k \in [1, \infty)$, and if the φ_k converge point-wise, as $k \to \infty$, to the continuous function φ on Ω with the coercive property $\varphi(x) \to \infty$ as $\|x\| \to \infty$, then $x^* \in \mathrm{argmin}_{\{x \in \Omega : g(x)=0\}} \varphi(x)$.*

Proof.

Existence: The set $\{x \in \Omega : g(x) = 0\}$ is non-empty by hypothesis. Let $\{x_n\}$ be a minimizing sequence in this set; that is, $\varphi(x_n)$ converges to the infimum. By the coercive property of φ, $\{x_n\}$ is bounded. Since Ω is closed, a subsequence converges to some $x' \in \Omega$. By the continuity of φ on Ω, $\varphi(x')$ equals the infimum, which is therefore finite and attained.

We proceed to show that x^* is a solution. Let $\tilde{x}$ be a solution of the problem $\min_{\{x \in \Omega : g(x)=0\}} \varphi(x)$; then for any $\epsilon > 0$, we have, for sufficiently large k, (since φ_k converges point-wise to φ)

$$\varphi_k(\tilde{x}) < \varphi(\tilde{x}) + \epsilon$$

Since x^* is a solution for φ_k, we also have

$$\varphi_k(x^*) \le \varphi_k(\tilde{x})$$

Combining the two gives

$$\varphi_k(x^*) \leq \varphi_k(\tilde{x}) < \varphi(\tilde{x}) + \epsilon$$

Or,

$$\varphi_k(x^*) \leq \varphi(\tilde{x}) + \epsilon$$

Taking the limit as $k \to \infty$ we have:

$$\varphi(x^*) \leq \varphi(\tilde{x}) + \epsilon$$

Since this is true for all $\epsilon > 0$ we necessarily have that

$$\varphi(x^*) \leq \varphi(\tilde{x})$$

Noting that x^* satisfies the constraint, $g(x) = 0$, we infer that x^* is also a solution.

Theorem 1. *The linear unbiased estimator previously obtained for the generalized least squares (best linear unbiased estimator) problem, $L^*\mathbf{b} = (A^T V^{-1} A)^{-1} A^T V^{-1}\mathbf{b}$, is also a solution that minimizes the maximum eigenvalue of the estimator's covariance matrix, LVL^T, subject to the constraint $L_{m\times n}A_{n\times m} = I_{m\times m}$. (We claim it is a minimizer; the spectral objective need not have a unique one.)*

Proof. We apply Lemma 1 with the variable $L \in \mathbf{R}^{m\times n}$ (identified with $\mathbf{R}^{mn}$), $\Omega = \{L \in \mathbf{R}^{m\times n} : L\,A = I\}$ (closed, and non-empty since $L^* \in \Omega$), $\varphi_k(L) = \sqrt[k]{\mathrm{tr}((LVL^T)^k)}$ (the k^{th} Schatten norm of LVL^T), and $\varphi(L) = \lambda_{\max}(LVL^T)$ (the spectral norm of LVL^T, which is positive definite since L has full row rank and V is positive definite). The constraint $g(L) = 0$ is vacuous since Ω already encodes $L\,A = I$. Theorem 1 follows after collecting the facts:

- For a positive definite matrix with eigenvalues $\lambda_i > 0$, $\sqrt[k]{\sum_i \lambda_i^k} \to \max_i \lambda_i$ as $k \to \infty$; so $\varphi_k \to \varphi$ point-wise.
- The L_2 matrix norm of a positive definite symmetric matrix is its maximum eigenvalue; φ is a continuous function of L.
- φ is coercive on Ω: since V is positive definite, $\lambda_{\max}(LVL^T) \geq \lambda_{\min}(V)\|L\|_2^2$, which grows without bound as $\|L\|$ grows.
- L^* is a *common* minimizer of φ_k for every $k \in [1, \infty)$: by Section 2, minimizing φ_k is the same as minimizing its k^{th} power $\mathrm{tr}((LVL^T)^k)$, whose unique minimizer on Ω is L^*.

References

[1] A. C. Aitken, On least squares and linear combination of observations, *Proceedings of the Royal Society of Edinburgh* **55** (1935), 42–48.

[2] C. R. Rao, *Linear Statistical Inference and Its Applications*, 2nd ed., Wiley, 1973.